INCH-POUND
MIL-DTL-965F(SH)
24 March 2020
SUPERSEDING
MIL-H-965E(SH)
22 December 1994

DETAIL SPECIFICATION

HEATERS, WATER, ELECTRIC, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes electric, self-contained, water heaters (see 6.5.2) used aboard United States naval surface ships and submarines as a complete system and ensures that all system components are compatible and properly integrated with regard to size, function, performance, endurance, safety test procedures, shock and vibration, and standardization.

1.2 Classification. Electric water heaters are of the following types and operating ranges, as specified (see 6.2).

1.2.1 Type. The types of electric water heaters are as follows:

a. Type ST – Storage tank (see 6.5.7).

b. Type IN – Instantaneous (also called tankless or demand-type) (see 6.5.3).

c. Type SI – Semi-instantaneous (see 6.5.6).

1.2.2 Operating ranges. Electric water heaters are used for several purposes aboard ship with setpoint temperatures as follows (degrees Fahrenheit [°F] and degrees Celsius [°C]):

a. Personnel use (showers and lavatories): 125 °F (51 °C)

b. Non-personnel use:

(1) Laundry and low temperature dishwashers: 140 °F (60 °C)

(2) Grease interceptor hoods: 160 °F (71 °C)

(3) Sanitizing scullery sinks: 170 °F (77 °C)

(4) High temperature booster: 180 °F (82 °C). High temperature booster heaters are only required to support specific installations such as older dishwasher models.

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1.3 Part or identifying number (PIN). The PINs to be used for electric water heaters acquired to this specification are created as follows:

M	965	-	XX	-	XX	-	XX	-	XXX
Prefix for Military Specification	Specification Number	-	Type: ST = storage tank IN = instantaneous SI = semi-instantaneous	-	Demand flow rate in gallons per minute (GPM) for type IN and SI heaters	-	Storage capacity in gallons for type ST and SI heaters	-	Power rating in kilowatts (kW)
Examples: PIN M965-ST-XX-25-005 is a 25-gallon storage tank, 5-kW electric water heater. PIN M965-SI-02-08-005 is a semi-instantaneous, 2 GPM, 8-gallon, 5-kW electric water heater.									

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-P-645 - Paint, Alkyd Type, Formula Number 84

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

MIL-DTL-12133 - Washer, Spring Tension, General Specification for

MIL-DTL-12133/4 - Washer, Spring Tension, Wave, CRES 302

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plates, Identification

MIL-DTL-15090 - Enamel, Equipment, Light Gray (Navy Formula No. 111)

MIL-DTL-18240 - Fastener Element, Self-Locking, Threaded Fastener, 250 °F Maximum

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MIL-G-18997	-	Gauge, Pressure, Dial Indicating
MIL-PRF-20042	-	Flanges, Pipe, Bronze (Silver Brazing)
MIL-H-22577	-	Heating Elements Electrical: Cartridge, Strip and Tubular Type
MIL-DTL-24696	-	Gasket, Sheet, Non-Asbestos
MIL-DTL-32613	-	Controller, Auxiliary-System, Naval Shipboard Use

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-769	-	Insulation Requirements for U.S. Naval Vessels
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-1472	-	Human Engineering
MIL-STD-1568	-	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons System
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices)
MIL-STD-2169	-	High-Altitude Electromagnetic Pulse (HEMP) Environment

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment
MIL-HDBK-263	-	Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)
MIL-HDBK-454	-	General Guidelines for Electronic Equipment
MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0900-LP-001-7000	-	Fabrication and Inspection of Brazed Piping Systems
S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

AIA/NASM 25027	-	Nut, Self-Locking, 250 °F, 450 °F, and 800 °F
NAS 411-1	-	Hazardous Materials Target List

(Copies of these documents are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

CSA/AM-ANSI Z21.22	-	Standard for Relief Valves and Automatic Gas Shutoff Devices for Hot Water Supply Systems

(Copies of this document are available online at http://webstore.ansi.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5	-	Pipe Flanges and Flanged Fittings
ASME B40.200	-	Thermometers, Direct Reading and Remote Reading
ASME BPVC Section IV	-	Rules for Construction of Heating Boilers

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ASME BPVC, Section VIII, Division 1 - Rules for Construction of Pressure Vessels

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

(Copies of this document are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10012 - Measurement Management Systems – Requirements for Measurement Processes and Measuring Equipment

(Copies of this document are available online at www.iso.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA HP 7 - Electrical and Electronic PVC, PVC/Nylon, and PE/Nylon 105 °C Hook-Up Wire, Types B, C, D, BN, CN, and DN (600, 1000, and 3000V), and Types J and JN 75 C (600V)

NEMA 250 - Standard for Enclosures for Electrical Equipment (1000 Volts Maximum)

(Copies of these documents are available online at www.nema.org.)

SAE INTERNATIONAL

SAE J1086 - Numbering Metals and Alloys

(Copies of this document are available online at www.sae.org.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.

3.2 Materials. Unless otherwise specified (see 6.2), materials used in manufacturing the heaters shall be as listed in table I. Material compatibility shall be in accordance with MIL-STD-889.

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TABLE I. Electric water heater system material list.

Part	Material [1/]
Element flange; tank shell; heads or cover plates; tank flanges; element support; all parts in with water	90-10 copper-nickel UNS C70600, C70620 70-30 copper-nickel UNS C71500, C71520 Copper UNS C12200 Stainless steel UNS S30400/S30403 [2/], S31600/S31603 Duplex stainless steel UNS S32101, S32205
Heating element flange; individual element flange fittings attached to the element sheath; element supports, spacer plates, rods	Steel 316L UNS S31603 Duplex stainless steel UNS S32101, S32205
Heating element sheaths	Nickel-chromium-iron alloy UNS N06625 (Inconel 625) MIL-H-22577 [3/]
Gaskets	MIL-DTL-24696
Jacket; fasteners	Corrosion-resistant metal or materials suitably protected against corrosion (see 3.2.2)
Thermal insulation	Fiberglass wool (MIL-STD-769)

NOTES:

1/ SAE J1086 Unified Numbering System (UNS) designation. Reference to a specific UNS designation permits the use of any one of the many alloy specifications and standards listed in SAE J1086 for that UNS number.

2/ Stainless 304 alloys shall not be used for wetted surfaces.

3/ Includes copper sheathed elements as permitted in MIL-H-22577 for certain existing in-service applications (including submarine potable water heaters) that currently use copper elements.

3.2.1 Prohibited materials. Heaters shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs. Lead and its compounds shall not be used where they can come in contact with freshwater. The lead content in bronze alloys shall not exceed levels allowed by ASTM B61.

3.2.2 Corrosion protection.

3.2.2.1 Corrosion-resisting materials. Corrosion-resisting steel, copper, bronze, copper-silicon, copper-nickel, copper-nickel alloys, or other specified materials (see 6.2) are considered corrosion-resisting materials. Where corrosion-resistant steel is used, it shall be type 304 or 316 (UNS S30400 or S31600) or duplex stainless steel UNS S32101 or, S32205. Type 304 shall not be used in contact with water. Corrosion-resistant steel, when fabricated by any method that tends to reduce corrosion-resisting properties, shall be normalized (see 6.5.5) to restore those properties before being used in the assembly of any heater.

3.2.2.2 Non-corrosion-resisting materials. Parts other than corrosion-resistant materials shall be protected against corrosion by the use of chemicals, electrolytic processes, plating, or paints and enamels. MIL-STD-1568 provides guidance.

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3.2.2.3 Hardware. Bolts, nuts, studs, pins, screws, and other such fastenings or fittings shall be of a corrosion-resistant material. Self-tapping screws (i.e., screws of a type that only part of the threading can cut, which means if the screw is extracted and reinserted, it will not cut new threads, but will follow the previous ones) with machine screw threads may be used in the cabinet assembly. Sheet metal screws with sheet metal threads shall not be permitted.

3.2.2.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction. Heaters shall be constructed as specified in 3.3.1 through 3.3.3.

3.3.1 General. The electric water heater system shall be a complete system in which all components and materials are compatible and properly integrated with regard to size, function, and performance. Heaters shall be of storage tank, instantaneous, or semi-instantaneous designs as specified (see 1.2). The heater shall heat and discharge water at the specified temperature and capacity (see 6.2) upon demand. A natural circulation system that entails the use of shrouds shall be employed to minimize water stratification. The heater system shall have accessibility to all internal and external parts for ease of serviceability. The location of inlet lines, outlet lines, and auxiliary components shall not obstruct flange connections and other field removable parts. In addition, water heaters shall be constructed so that the tank (if used) or heating chamber tubing can be cleaned, and the heating elements and thermostatic switch removed without disturbing the insulation on the heater or removing any piping. Auxiliary water heater system components (relief valve, heating elements, thermometer, drain connection, drain valve, and controller or magnetic contactor) shall be installed to provide accessibility for inspection, tool maneuverability, and removal or repair of these components. See figure 1 for typical electric water heater systems. Water pressure loss through the heater shall be the minimum attainable.

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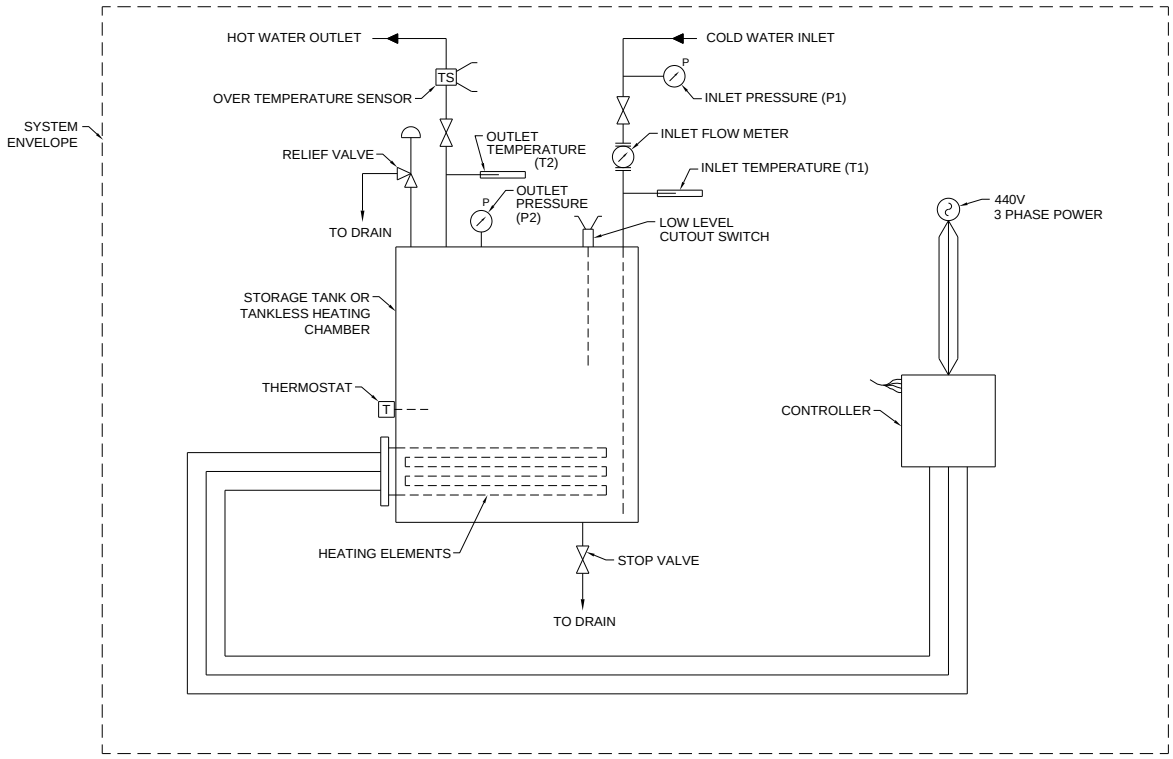


FIGURE 1. Typical electric water heater system.

3.3.2 Capacity. The load (see 6.5.4) capacity for type ST heaters shall be specified as the tank liquid volume in gallons. The load capacity for type SI heaters shall be specified in GPM of output water and storage size. The load capacity for type IN heaters shall be specified in GPM of output water (see 6.2).

3.3.3 Response time. The attainable continuous flow rate of heated water, after initial tank depletion, shall be 1 hour or as specified (see 6.2). The response time for an instantaneous or semi-instantaneous heater to continuously produce its rated flow rate (in GPM) of hot water (125 °F [51 °C]) shall be not greater than 40 seconds when supplied with 40 °F (4 °C) inlet water.

3.4 Reliability and maintainability. Each heater system shall have an operating life of 100,000 hours. Human factors requirements of MIL-STD-1472 shall be met. Components in the system shall be free of obstruction when requiring servicing or replacement and be constructed to be operated and maintained by two seamen with training equivalent to a competent journeyman with skills in plumbing and electrical.

a. Mean Time Between Failures (MTBF): 10,000 hours of operation. The MTBF shall be supported by analysis or industry information provided and verified by the producer. MIL-HDBK-217 predictive methods may be used.

b. Mean Time to Repair (MTTR): 2 hours.

c. Mean Time Between Corrective Maintenance (MTBCM): 1,000 hours. No more than 2 hours of corrective maintenance shall be necessary per 1,000 hours of operation.

d. Operational Availability (A_o): 0.95.

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3.5 Performance conditions. Unless otherwise specified (see 6.2), heater systems shall meet the specified performance requirements under the following parameters:

a. Heater inclination: shall operate properly under permanent 15-degree inclination in any direction and symmetrical rolling to 45-degree inclination in any direction.

b. Inlet water temperature: 40 °F to 90 °F (4 °C to 33 °C).

c. Inlet and operating water pressure: 30 to 150 pounds per square inch (lb/in^2) (2 to 10.3 bar) gauge.

d. The operating voltage shall be one of the following: 440 volts alternating current (VAC), 115 VAC, special AC voltages, 115 volts direct current (VDC), 230 VDC, or as specified (see 6.2). For AC voltages including special AC voltages, equipment shall be 60 hertz and in accordance with MIL-STD-1399-300-1. For DC voltages, the element shall operate satisfactorily on voltages up to 105 percent of the specified rating. When heating elements are intended for operation on a voltage other than line voltage (by series or wye connections), the line-to-line voltage shall also be indicated. The heating element shall be insulated for the line voltage and shall have electrical creepage and clearance dimensions for line voltage in accordance with MIL-DTL-917. Unless otherwise specified (see 6.2), control devices shall operate on the same voltage as the heating elements.

e. Outlet water temperature and over-temperature (see 3.9.5) setpoints:

(1) Personnel use:	125 °F (51 °C)	Over-temperature at 140 °F (60 °C)
(2) Non-personnel use:		
(a) Laundry and low temperature dishwash:	140 °F (60 °C)	Over-temperature at 150 °F (66 °C)
(b) Grease interceptor hoods:	160 °F (71 °C)	Over-temperature at 170 °F (77 °C)
(c) Sanitizing scullery sinks:	170 °F (77 °C)	Over-temperature at 180 °F (82 °C)
(d) Dishwash high temperature booster:	180 °F (82 °C)	Over-temperature at 190 °F (88 °C)

f. Steady-state performance: water heater systems shall maintain steady-state performance at all conditions of operation.

g. Hydrostatic withstand: design pressure for heaters shall be the ASME 150 lb/in^2 (10.3 bar) rating.

3.6 Temperature adjustments range. Heater operating ranges shall be adjustable, at minimum, to operate between the following temperature ranges: personnel use, 90 °F and 140 °F (32 °C and 60 °C); non-personnel use, 90 °F and 180 °F (32 °C and 82 °C).

3.7 Fabrication. All heaters shall be welded in accordance with S9074-AQ-GIB-010/248 and S9074-AR-GIB-010/278 for class A-4 pressure vessels or brazed in accordance with 0900-LP-001-7000. Storage tank water heater systems shall be in accordance with the requirements of ASME BPVC Section IV or Section VIII as appropriate to the design. Unless otherwise specified (see 6.2), the shell shall be capable of a working pressure of not less than 150 lb/in^2 (10.3 bar) gauge. Storage tank water heaters shall consist of a pressure tank that shall be insulated against heat loss and protected by a jacket. Thermal insulation and lagging shall be in accordance with MIL-STD-769. Insulation and lagging shall not interfere with removal of the heater elements or thermostatic switches from the tank or the heating chamber for instantaneous and semi-instantaneous heaters.

3.7.1 Tanks and heating chambers. Tanks of storage tank heaters and, when applicable, heating chambers of instantaneous and semi-instantaneous heaters shall consist of a shell with heads or cover plates that shall be welded or silver brazed as specified in 3.7.6. The tank shall be furnished with a base support that is either welded, silver brazed, or bolted to the tank. The base support shall be deck mounted by a minimum of four bolts measuring ⅝inch, 11 UNC (M18x2).

3.7.1.1 Shells. Shells shall be cylindrical; fabricated from sheet, piping, or tubing material, as specified in table I; and either silver brazed or seam welded in accordance with 3.7.6. A means for internally supporting the heating element(s) during field removal and reinsertion shall be provided. A support that secures the heating element(s) in both horizontal and vertical directions shall be provided opposite to the heating element mounting flange(s).

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3.7.1.2 Heads or cover plates. Heads or cover plates shall be either cast, forged, or formed and, when applicable, removable by bolting at the trunk shell flange. The head or cover plate material shall be as specified in table I.

3.7.1.3 Relief valve bosses. Heaters shall be equipped with a boss for a relief valve. Connections for the relief valve's boss shall be a threaded union or flanged connection in accordance with MIL-STD-777 (see 3.9.3).

3.7.2 Jackets. Jackets shall be used to enclose the tank and insulating material. The jacket material shall be as specified in table I.

3.7.3 Flanges. Flanges shall be welded to the tank and shall be class 150 in accordance with ASME B16.5; ASME BPVC Section VIII, Division 1, Appendix 2; or MIL-PRF-20042.

3.7.4 Heating elements. Heating elements shall be in accordance with MIL-H-22577, type III. The heating element sheath shall be secured to the flange or individual element flange adapter fitting by welding. Welding shall be in accordance with S9074-AR-GIB-010/278 and S9074-AQ-GIB-010/248.

3.7.4.1 Heating element flanges. Heating element flanges, fittings, supports, and spacer rods shall be of material as specified in table I.

3.7.4.2 Heating element sheaths. Heating element sheaths shall be seamless or welded nickel-chromium-iron alloy UNS N06625 (Inconel 625) or other material in accordance with MIL-H-22577. Heating element sheath material shall be bright annealed after all fabrication steps up to insertion into the flange or individual element flange fitting. The heating element terminal ends shall be hermetically sealed using the ceramic to metal type seal as specified in MIL-H-22577. In lieu of lock washers, Belleville washers in accordance with MIL-DTL-12133/4, self-locking nuts in accordance with AIA/NASM 25027 or MIL-DTL-18240, or double locking nuts may be used for terminal connections.

3.7.4.3 Heating element wattage. The wattage rating shall be selected to give the energy required to meet the response time (see 3.3.3). Storage tank heaters' and semi-instantaneous heaters' watt density shall be not greater than 50 watts per square inch (0.08 watt per square millimeter) of element heating surface. Instantaneous heaters' watt density may be as required for heated flow rate, not exceeding 185 watts per square inch (0.3 watt per square millimeter).

3.7.5 End connections. End connections for the water inlet and outlet shall terminate in take-down flanges in accordance with ASME B16.5, class 150 or MIL-PRF-20042. Union fittings in accordance with MIL-STD-777 may be used where applicable. Threaded pipe connections shall not be used. Heaters shall be equipped with necessary drain valves to permit complete drainage of the heater system.

3.7.6 Welding and silver brazing. Welding and silver brazing shall be in accordance with S9074-AQ-GIB-010/248, S9074-AR-GIB-010/278, or 0900-LP-001-7000, as appropriate.

3.7.7 Heater supports. The heater systems shall include a frame to support and elevate all heater components to a height permitting all connections to be easily accessible without having to remove the framework or other obstructions (see 3.3.1).

3.8 Identification and information plates. Heaters shall have an identification plate mounted on the terminal box or on the tank in full view, and it shall contain the following data:

a. "ELECTRIC WATER HEATER" (in capital letters).
b. Manufacturer's name and address.
c. Manufacturer's model, serial, and part numbers.
d. Type and capacity.
e. Contract or purchase order number.
f. National stock number.
g. Voltage, wattage, phases, and amperage.
h. Design outlet temperature.

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i. Maximum outlet temperature.

Identification plates shall be in accordance with type C of MIL-DTL-15024 and MIL-P-15024/5. Each controller shall include a schematic diagram in accordance with MIL-DTL-2212. The diagram shall be secured inside the controller in a location that is readily accessible.

3.9 Controls, safety devices, and indicators. Controls, safety devices, and indicators as shown on figure 1 shall be furnished as part of the water heater system.

3.9.1 Temperature sensing devices. Temperature sensing devices shall be provided with each heater and shall be of the types specified (see 6.2). The same type of sensor used to input to the controller shall be used to input to alarms, controls, and protective devices and, unless otherwise specified (see 6.2), shall maintain the outlet temperature to ±5 °F (±2 °C). Thermostatic switches shall conform to MIL-DTL-2212 and be wired separately from the over-temperature safety device and the low-level cutout switch.

3.9.2 Contactors and controllers. A magnetic contactor or cybersecure controller in accordance with MIL-DTL-32613 shall be furnished by the heater manufacturer with each water heater as specified (see 6.2). The contactor or controller shall be mounted in a totally enclosed, drip-proof, corrosion-resistant steel enclosure in accordance with MIL-DTL-2036 and shall have a removable, corrosion-resistant steel cover for access to the control devices. In addition, the enclosure mounting shall be in an area to prevent heat from the tank to exceed the operating temperature limits of the control devices. In the case where the controller requires fan cooling, a spray-tight enclosure in accordance with NEMA 250 shall be used. Power leads shall not enter the enclosure through the cover plate. Nuts, bolts, and screws shall be copper-nickel or corrosion-resistant steel. The enclosure gasket shall be made of rubber.

3.9.3 Temperature and pressure (T&P) relief valve. Heaters shall have a T&P relief valve with a lifting device constructed and installed in accordance with the ASME BPVC Section IV and CSA/AM-ANSI Z21.22, with exception to end connections. The relief valve shall have male thread union ends in accordance with MIL-DTL-1183 with back-up retaining rings and O-rings installed, or flanges in accordance with MIL-STD-777 and, unless otherwise specified (see 6.2), set to relieve at the design pressure rating. Rupture disks shall not be used as a means of preventing over-pressurization.

3.9.4 Temperature and pressure gauges. Heaters shall have the temperature gauges installed in the inlet and outlet lines. Each heater with a tank or reservoir shall have a pressure gauge located on top of the tank or reservoir. Each temperature gauge shall have a direct reading 3-inch (76-mm) or larger dial size in accordance with ASME B40.200. Similarly, each pressure gauge shall have a direct reading 4½-inch (114-mm) or larger dial size in accordance with MIL-G-18997. The temperature gauges shall have a range from 20 °F to 240 °F (-7 °C to 116 °C). Pressure gauge ranges for the supply water line and tank shall be 0 to 200 lb/in² (0 to 14 bar) gauge in accordance with MIL-G-18997. Thermowells shall be socket welded, socket brazed, or threaded ¾-inch NPS (19-mm) in accordance with ASME B40.200.

3.9.5 Over-temperature safety device. Heaters shall have an adjustable over-temperature safety device in the tank top or outlet water line in the event that the outlet water temperature rises above the heater temperature setting. Installation of a T&P relief valve does not suffice to meet this requirement. The safety device shall trip at the preset maximum temperature (see 3.5.e) and shall deenergize the heating element(s). The safety device shall simultaneously activate a local visual alarm (red telltale light) to alert personnel to an over-temperature condition. The safety switch shall have a manual reset feature that shall be utilized to energize the heating element(s) once a safety shutdown occurs.

3.9.6 Low-level cutout device. Heaters with tanks shall have a low-level cutout electrical device installed in the event of a loss of supply water. The low-level cutout electrical device shall be set to deenergize the heating element(s) to prevent overheating. The low-level cutout device shall simultaneously activate a visual alarm (red telltale light) to alert personnel of the loss of water supply. In addition, the device shall have a manual reset feature that shall be utilized to energize the heating element(s) once a safety shutdown occurs.

3.9.7 Flow sensor. Instantaneous and semi-instantaneous heaters shall be furnished with a flow sensor on the water inlet that will activate the heating elements as necessary.

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3.9.8 Wiring. Wire used to connect the electrical components shall be in accordance with NEMA HP 7. Internal wiring shall be completed to an approval terminal enclosure for interconnection of each component for field installation. Wire shall be sized to suit load requirements. Wires more than 12 inches (304 mm) in length shall be marked for identification. Marking shall be in accordance with MIL-DTL-917.

3.10 Electrical equipment. Electrical equipment shall be in accordance with the applicable requirements of MIL-DTL-917. DC water heaters shall use DC electrical accessories. For DC water heaters with AC accessories, a warning label plate in accordance with MIL-DTL-2212 shall be mounted on the front cover of the enclosure to warn service personnel that there are two sources of power to the enclosure. Unless otherwise specified (see 6.2), electrical equipment shall be suitable for the power supply as stated above (see 3.5.d).

3.11 Electromagnetic emission and susceptibility. The monitoring system to enable the system shall be in accordance with the requirements of MIL-STD-1399-300-1 and demonstrate compliance in accordance with the general requirements of MIL-STD-461 and emission and susceptibility requirements and limits CE101, CE102, CS101, CS114, CS116, RE101, RE102, and RS103 (see 4.5.16).

3.11.1 Electromagnetic pulse (EMP) protection. Control systems as specified herein shall survive and operate after being subjected to an EMP environment in accordance with MIL-STD-2169.

3.11.2 Electrostatic discharge (ESD). Control systems as specified herein shall be in accordance with the ESD requirements of MIL-STD-1686. MIL-HDBK-454 and MIL-HDBK-263 provide guidance.

3.12 Safety. Equipment shall be constructed to ensure safety to operating and maintenance personnel in accordance with MIL-DTL-917 and MIL-STD-1472. Heater components and piping shall be insulated to limit the surface temperature to a maximum of 125 °F (51 °C) when subjected to the maximum temperature possible during operation. Thermal insulation shall be in accordance with MIL-STD-769. When specified (see 6.2) for personnel use heaters, an International Association of Plumbing and Mechanical Officials (ASSE) approved anti-tampering device to prevent unauthorized personnel from adjusting the temperature controls shall be provided.

3.13 Dielectric withstand and insulation resistance. Heater shall withstand the dielectric strength and insulation resistance tests as specified in 4.5.12.

3.14 Special tools. Special tools required for servicing the heater and associated valves and components shall be as specified (see 6.2). Special tools are defined as tools not listed in the GSA Global Supply Catalog.

3.15 Painting. Unprotected external surfaces of non-corrosion-resistant ferrous parts shall be treated as follows:

a. Sandblasted clean.

b. Primed with one coat of primer in accordance with TT-P-645.

c. Painted with one coat of enamel in accordance with MIL-DTL-15090.

3.16 Shock and vibration. Heater systems shall be in accordance with the grade B, deck mounted, class I, type A shock requirements of MIL-DTL-901 and the type I vibration requirements of MIL-STD-167-1. The tests shall be conducted at the heater operating temperatures and pressures. The tested system shall include all components required to make a complete and operable heater system as shown on figure 1.

3.17 Limiting dimensions. Length, width, height, weight, and element withdrawal dimensions shall be as specified (see 6.2).

3.18 Cleaning kits. When specified (see 6.2), a cleaning kit to remove scale buildup due to calcium deposits shall be furnished.

3.19 Workmanship. Heaters shall be free of visual and operating defects such as sharp edges, dents, scratches, tears, paint runs, and malfunctioning controls. Mating surfaces of flanges and mounting pads of supports shall be machined true.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the first heater of each new design when a first article sample is required (see 3.1). This inspection shall include the examinations and tests specified in table II. Some tests may be able to be conducted concurrently; however, the shock test, inclined operation test, and vibration test shall be performed separately.

4.3 Conformance inspection. Conformance inspection of lot samples shall include the examinations and tests specified in table II.

4.3.1 Lot. Heaters of the same use, type, and size offered for delivery within a time span not greater than 1 year shall be considered a lot.

4.3.2 Sampling for examination and tests. A random sample of heaters shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.4 Instrumentation calibration. Measuring and test equipment used in the performance of tests shall be calibrated in accordance with ISO 10012.

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TABLE II. First article and conformance inspections.

Characteristic	Requirement	First article	Conformance
Classification and capacity	1.3, 3.3.2	4.5.1	4.5.1
Materials	3.2	4.5.1	---
Prohibited materials	3.2.1	4.5.2	---
Construction	3.3, 3.3.1	4.5.1	4.5.1
Response	3.3.3	4.5.7	---
Reliability and maintainability	3.4	4.5.10	---
Performance: inclination	3.5.a	4.5.6	---
Performance: inlet water	3.5.b	4.5.7	---
Performance: inlet and operating water pressure	3.5.c	4.5.7	---
Performance: voltage	3.5.d	4.5.14	---
Performance: outlet water temperature	3.5.e	4.5.7 4.5.7.1	4.5.7
Performance: steady-state operation (type ST heaters only)	3.5.f	4.5.15	4.5.15
Performance: hydrostatic withstand	3.5.g	4.5.3	4.5.3
Temperature adjustment range	3.6	4.5.1	---
Fabrication	3.7, 3.7.1.3, 3.7.5-3.7.7	4.5.1	---
Fabrication: heating elements	3.7.4	4.5.7	---
Identification plates	3.8	4.5.1	---
Controls	3.9	4.5.8, 4.5.9, 4.5.11, 4.5.13, 4.5.14	---
Electrical	3.10	4.5.1	---
Electromagnetic	3.11, 3.11.1, 3.11.2	4.5.16	---
Safety	3.12	4.5.1	---
Dielectric and insulation resistance	3.13	4.5.12	4.5.12
Special tools and cleaning kits	3.14, 3.18	4.5.1	---
Painting	3.15	4.5.1	---
Shock and vibration	3.16	4.5.4, 4.5.5	---
Limiting dimensions	3.17	4.5.1	---
Workmanship	3.19	4.5.1	4.5.1

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4.5 Examinations and tests.

4.5.1 Heater system visual and dimensional examinations. Heaters shall be examined and measured to verify compliance with the requirements of materials, adjustment, fits, leaks, workmanship, and design requirements of this specification, to include the requirements of table III. Any defects found during inspection as outlined in table III shall be cause for rejection.

a. The external fittings shall be properly secured.

b. Bolts, nuts, and screws shall be tight; equipment and accessories shall be properly fastened and secured.

c. No parts shall be fractured, split, torn, dented, or otherwise damaged to adversely affect serviceability.

d. There shall be no sharp or ragged edges on any surfaces that may be injurious to personnel.

e. The temperature control shall be properly set and functioning.

TABLE III. Visual and dimensional examinations.

Visual and dimensional aspects	Requirements
Classification	1.2
Materials	3.2
Capacity	3.3.2
Voltages	3.5.d
Temperature adjustments	3.6
Tanks	3.7.1
Shells	3.7.1.1
Heads or cover plates	3.7.1.2
Jackets	3.7.2
Flanges	3.7.3
Heating elements	3.7.4
End connections	3.7.5
Welding and silver brazing	3.7.6
Heater supports	3.7.7
Identification and information plates	3.8
Controls	3.9
Thermostatic switches	3.9.1
Contactor or controller	3.9.2
T&P relief valve	3.9.3
Temperature and pressure gauges	3.9.4
Over-temperature safety device	3.9.5
Low-level cutout device	3.9.6
Wiring	3.9.8
Safety and thermal insulation	3.12
Special tools	3.14
Painting	3.15
Limiting dimensions	3.17

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TABLE III. Visual and dimensional inspections – Continued.

Cleaning kits	3.18
Workmanship	3.19

4.5.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.1 and 6.2).

4.5.3 Hydrostatic testing. Heat transfer elements including vessels, shells, heads, cover plates, and end connections shall be hydrostatically tested for design pressure of 150 lb/in² (10.3 bar) in accordance with ASME BPVC Sections IV or VIII as appropriate to the design. The heater shall be connected for testing as required to simulate shipboard operation. Gauges, thermometers, and controls shall be calibrated before the test to manufacturer specification and traceable to the National Institute of Standards and Technology (see 4.4). Temperature and pressure gauges shall be isolated during the test. The pressure shall be isolated and maintained in the heaters for a minimum of 30 minutes. Hydrostatic testing shall be completed before the shock test is performed (see 4.5.4). The failure to maintain specified ASME BPVC hydrostatic test pressure in a heater for 30 minutes, or visible evidence of leakage, permanent distortion, or collapse, shall be cause for rejection of the heater.

4.5.4 Shock test. Hydrostatic testing (see 4.5.3) shall be completed before the shock test is performed. Heater systems shall be tested in accordance with the grade B, deck mounted, class I, type A shock requirements of MIL-DTL-901. Upon completion of the test, the heater system shall be tested as specified in 4.5.3. Shock testing of a water heater shall be considered acceptable unless the following conditions occurred during the test or were evident at the conclusion of the test:

a. Fragmentation or missile effects of parts.

b. Displacement or dislodgement of any piece, part, or component in such a manner that would be hazardous to personnel in the vicinity of the water heater system or cause an electrical short circuit.

c. Shearing or tearing of metallic parts that would create projections or sharp edges.

d. Any potable water (hot or cold) leakage.

e. Failure to operate as specified in 1.2, 3.9.1, 3.9.5, 3.9.6, and 4.5.12.

f. Failure to meet requirements as specified in 4.5.3.

g. Post shock tear down inspection reveals internal damage preventing proper operation in accordance with (a) through (f) preceding. Post shock tear down inspection can be conducted as part of the maintainability demonstration (see 4.5.10.1).

Momentary opening and closing of contactor and relay contacts under shock is permitted when operation of the heater is not affected.

4.5.5 Vibration test. The environmental vibration test shall be conducted in accordance with MIL-STD-167-1 for type I equipment. Acceptance of the water heater system shall be in accordance with MIL-STD-167-1.

4.5.6 Heater inclination. Water heaters as specified in 3.5.a shall be constructed to operate properly under permanent 15-degree inclination in any direction and symmetrical rolling to 45-degree inclination in any direction for at least 1 hour. The established performance characteristics shall be monitored at 10-minute intervals throughout the test. The water heater system shall be acceptable if the required temperatures do not fluctuate beyond the limits specified in table I and no leakage of fluids occur.

4.5.7 Response continuous flow test. Each water heater shall be bench tested at its full continuous flow rate to ensure the required outlet water temperatures are as specified in 3.3.3. The test shall be conducted with 40 °F (4 °C) inlet water. If 40 °F (4 °C) supply water is not available, the test may be conducted with available inlet water and setpoint raised to the same temperature rise (delta-T) as would be required with 40 °F (4 °C) inlet water. The heater shall produce its rated flow rate at its rated output temperature and pressure for 2 hours continuously. Failure of the water heater to comply with 3.3.3 or 3.5 shall be cause for rejection.

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4.5.7.1 First article operational test for instantaneous and semi-instantaneous heaters. Type IN and SI new design heaters and controllers shall demonstrate heater and controller operational performance. Partial, full, and intermittent load testing of the type IN and SI heaters shall be conducted as specified below. Type SI heaters shall maintain a minimum recirculation rate of 10 to 15 GPM. The system demand (through) flow shall be adjusted as required to obtain the desired load for the ambient water temperature. For type IN heaters, this demand flow shall not exceed the rated design flow.

a. Partial to full load testing shall be conducted in one continuous time period consisting of step increases of 20 percent of the type SI or IN heaters rated capacity. The system demand (through) flow shall be adjusted as required to obtain the desired percent load. Each step shall be held for 60 seconds until full load is attained. Full load shall be held for 10 minutes. After 10 minutes at full load, the system demand (through) flow shall be decreased as required to obtain a 20 percent load decrease. Each step shall be held for 60 seconds until zero load is attained. Outlet water temperature shall be maintained at the desired setpoint within the specified tolerance. Failure of the heater to meet the specified conditions shall be cause for rejection.

b. Intermittent load testing shall be conducted in one continuous time period consisting of a series of shorter periods in accordance with table IV in which specified load conditions apply. The system demand (through) flow shall be adjusted as required to obtain the desired percent load. All normal operating conditions shall be maintained throughout the test. Outlet water temperature shall be maintained at the desired setpoint within the specified tolerance. At no point during this test shall a safety shut down device secure the heater due to an out tolerance condition. Failure of the heater to meet the specified conditions shall be cause for rejection.

TABLE IV. Type IN and SI heater operational test.

Heater load (kW)	Duration (min)	Recirculation flow (GPM) – type SI only	System demand flow (GPM)
0	30	10 to 15	0
100 percent	30	10 to 15	100 percent or as specified for type SI
5 percent	10	10 to 15	100 percent or as specified for type SI
10 percent	5	10 to 15	100 percent or as specified for type SI
20 percent	5	10 to 15	100 percent or as specified for type SI
0	10	10 to 15	0
40 percent	2	10 to 15	100 percent or as specified for type SI
0	60	10 to 15	0
60 percent	2	10 to 15	100 percent or as specified for type SI
0	20	10 to 15	0
80 percent	2	10 to 15	100 percent or as specified for type SI
0	2	10 to 15	0
100 percent	5	10 to 15	100 percent or as specified for type SI

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4.5.8 Low-level cutout device test. The low-level cutout device shall be tested for type ST and SI heaters. The low water safety device shall be tested with the heater operating at 5 percent load and temporarily modifying the low water monitoring circuit in order to simulate a low-level condition on the low-level safety device. Operation of the low-level safety device shall be demonstrated by manually draining the water from the heater storage. Provisions shall be made to temporarily monitor the water level heater shell and demonstrate that the low-level device actuates before the level drops below the top of the upper element sheath. The heater low-level cutout device shall open the circuit and de-energize the heating element when it is not completely submerged in the water. A red telltale light or other alerting indicator shall illuminate to indicate de-energization of the heating element has occurred. Failure of the low-level cutout device as specified in 3.9.6 shall be cause for rejection.

4.5.9 Over-temperature safety device test. The heater system over-temperature safety device tests shall be conducted at 5 percent load and full load for each of the circulation heaters and the associated controller that make up the complete system. This test shall be accomplished with a minimum recirculation flow of 10 to 15 GPM on type SI heaters. The system demand (through) flow shall be adjusted as required to obtain the desired load for the ambient water temperature. For type IN heaters, this demand flow shall not exceed the rated design flow. The over-temperature tests shall be accomplished by raising the temperature control setpoint on the controller until the heater secures on the high outlet water temperature alarm. This test shall be accomplished for each heater element. The outlet water temperatures shall be monitored not more than 12 inches (305 mm) downstream of the outlet water tank fittings. The test shall be repeated to verify operation of the sheath over-temperature safety device. For the purpose of this test, the heater sheath temperature alarm setpoint may be reduced to a temperature that can easily be attained by adjusting the control setpoint. A red telltale light or other alerting indicator shall illuminate to indicate de-energization of the heating element has occurred. Failure of the safety device to de-energize the heating element as specified herein shall be cause for rejection.

4.5.10 Maintainability and reliability.

4.5.10.1 Maintainability demonstration. The heater shall be disassembled and all components shall be reassembled and replaced. The controls, instruments, heating elements, relief valve, and low-level cutout device specifically shall be removed and replaced. The demonstration shall show that access and replacement of components is possible without personnel hazard and in accordance with procedures outlined in the technical manual. The personnel and specific types of tools used in the demonstration shall be identified. In addition, the demonstration shall verify that the MTTR requirement, as specified in 3.4, is satisfied by the design and construction of the heater assembly. The demonstration shall be conducted utilizing the technical manual and the recommended tools without the use of an expert mechanic. The demonstration procedures in MIL-HDBK-470 may be used. Particular attention shall be directed toward the possibility of loss of small parts by the maintenance crew.

4.5.10.2 Reliability. The heater shall meet the requirements for reliability as specified in 3.4. The validation procedures in MIL-HDBK-217 may be used.

4.5.10.3 Acceptance criteria. Failure of the system to be disassembled and reassembled, without personnel hazard, in accordance with procedures outlined in the technical manual shall be cause for rejection.

4.5.11 Thermostatic switch calibration. Calibration of the thermostatic switch shall be checked by a controlled-temperature liquid-bath system or by thermocouples attached to the sensing bulb of the temperature control to indicate the temperature at which the switch contacts open at various settings within the specified range. Failure of the switch contacts to open at various settings within the specified range shall be cause for rejection.

4.5.12 Dielectric and insulation resistance tests. These tests shall be made after all other first article inspections have been completed. Insulation between any pair of mutually insulated electrical circuits or between any electrical terminal and the unit's frame/base shall withstand the following conditions:

a. Insulation resistance not less than 10 megohms with 500+50/-0 VDC applied.

b. 60 hertz dielectric voltage of 1,500 volts root mean square (rms) or twice the rated voltage plus 1,000 volts rms, whichever is greater, for 1 minute without any arcing, sparking, or current leakage in excess of 10 milliamperes.

c. Hot and cold insulation resistance tests shall be performed on heating element assemblies after all welding and allied processes have been completed. The insulation tests shall be in accordance with MIL-H-22577.

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4.5.12.1 Measurement of test voltage. The measurement of the voltage used in dielectric tests shall be made by means of a voltmeter whereby the meter derives its voltage directly from the high voltage circuit, either by means of a voltmeter coil placed in the testing transformer or through an auxiliary potential transformer.

4.5.12.2 Points of application. The test voltage shall be successfully applied between each electrical circuit and the frame with all other electric circuits and metal parts grounded.

4.5.12.3 Acceptance criteria. The heater shall be able to withstand the dielectric tests without electrical breakdown such as corona, flashover, spark over (surface discharge), or breakdown (puncture).

4.5.13 Thermostat test. The heater shall be filled with water at 40 °F (4 °C), the temperature taken, and the thermostatic switch set to cut off at setpoint temperature (see 3.5.e). If 40 °F (4 °C) supply water is not available, the test may be conducted with available inlet water and setpoint raised to the same temperature rise (delta-T) as would be required with 40 °F (4 °C) inlet water. Rated voltage shall be applied to the heating circuit and the water heated until the thermostatic switch automatically cuts off. Failure of the thermostat to automatically cut off at the present temperature shall be cause for rejection.

4.5.14 Voltage temperature test. The control equipment shall be subjected to a 4-hour operation at the voltage specified to determine that the temperature rise of the current-carrying parts does not exceed the maximum values permitted in accordance with MIL-DTL-2212. Failure of the current-carrying parts to be in accordance with MIL-DTL-2212 shall be cause for rejection.

4.5.15 Storage tank steady-state performance test. The test shall consist of the monitoring of the heater system's temperature gauges to determine the temperature swing while the system is operating with 40 °F (4 °C) inlet water. If 40 °F (4 °C) supply water is not available, the test may be conducted with available inlet water and setpoint raised to the same temperature rise (delta-T) as would be required with 40 °F (4 °C) inlet water. The temperatures shall be monitored for a period of 1 hour with readings taken at 5-minute intervals. After 1 hour, approximately 25 percent of the delivered water shall be drawn from the tank. Temperature readings shall be monitored again for a period of 1 hour at 5-minute intervals. No adjustments shall be made to the heater during the steady-state performance tests other than as specified herein. Failure of the heater system to maintain a temperature swing of ±5 °F (±2 °C) shall be cause for rejection.

4.5.16 Electromagnetic interference. Compliance with the requirements of 3.11 shall be demonstrated by performing and passing electromagnetic emission and susceptibility tests in accordance with MIL-STD-461. Failure of the heater system to meet the requirements of 3.11 shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Electric water heater systems are to be used onboard naval ships where continuous hot water generation is required and military environmental conditions may be expected. Current stock of shipboard water heaters may be used until depleted; future requisitions for shipboard water heaters should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and use of heater required (see 1.2).

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c. When first article inspection is required (see 3.1).

d. Materials required if not as specified herein (see 3.2).

e. Requirement for prohibited materials verification (see 3.2.1 and 4.5.2).

f. Capacity (for storage tank heaters) and continuous flow rating (see 6.5.1) (all heaters) (see 3.3.1, 3.3.2, and 3.3.3).

g. Performance requirements if other than specified herein (see 3.5).

h. Operating and control device voltage if other than specified herein (see 3.5.d).

i. Working pressure if other than as specified (see 3.7).

j. Type of temperature sensing device required (see 3.9.1)

k. Temperature range adjustability if other than specified herein (see 3.9.1).

l. Contactor or cybersecure controller to be furnished (see 3.9.2).

m. Relief valve set pressure if other than as specified (see 3.9.3).

n Electrical power supply requirements if other than as specified (see 3.10).

o. Anti-tampering device if required (see 3.12).

p. Special tools to be furnished by the contractor if required (see 3.14).

q. Limiting dimensions (see 3.17).

r. Cleaning kits if required (see 3.18).

s. Sampling AQL (normally 1.0) (see 4.3.2).

t. Packaging requirements (see 5.1).

u. Recommended documentation (see 6.3).

6.3 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition (see 6.2) of any electric water heater under this specification:

a. Engineering drawings.

b. Materials listing.

c. Installation drawings.

d. Technical manual in accordance with MIL-DTL-24784.

e. Visual aids and maintenance cards in accordance with MIL-STD-3034.

f. Test reports.

g. Certifications.

h. Onboard repair parts list. Table V has minimum suggestions.

TABLE V. Onboard repair parts.

Name of part	Quantity based on number of heaters installed		
	1-5	6-10	11-20
Heater element assembly	1	1	2
Thermostatic switch assembly	1	2	4
Gaskets and seals	100 percent	100 percent	100 percent
Contactor or controller (in accordance with MIL-DTL-32613)	As applicable		

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Continuous flow rating. A measure of a water heater's capacity to continuously supply a volume of heated water. For non-storage heaters, this is a GPM rate. For storage heaters, this is a GPM rate after an initial depletion, usually around 70 percent of tank capacity, that can be sustained indefinitely.

6.5.2 Heater. Used synonymously with "heater system" within this specification to mean the complete assembly of heat exchanger, control, monitoring, and miscellaneous equipment supplied.

6.5.3 Instantaneous water heater. A heater in which the power input (kW) is sufficient to heat the incoming water based on inlet temperature and flow rate to the desired outlet temperature and which energizes upon demand. These typically have an internal volume of 1 gallon or less and are installed at the point of use. Typically, these heaters do not have relief valves or manually resetting high-limit thermostats.

6.5.4 Load. Refers to the demand for flow of outlet water. A full load condition is when the heater is required to deliver the rated capacity.

6.5.5 Normalized. A heat treatment process in which the steel is heated to 80 °F to 122 °F (30 °C to 50 °C) above the critical temperature and cooled in air for a suitable time. The main difference between normalized and annealed metals is that the cooling rate of normalizing is a little faster, so the production cycle of normalizing is shorter.

6.5.6 Semi-instantaneous water heater. A heater in which the power input (kW) is sufficient to heat the incoming water based on inlet temperature and flow rate to the desired outlet temperature and has some storage volume. These heaters energize upon demand and as needed to maintain the stored water at the set temperature, and they are always equipped with relief valves. These are typically circulation-loop heaters that bridge a gap between instantaneous and storage type water heaters.

6.5.7 Storage tank water heater. A heater in which the power input (kW) is not sufficient to heat the incoming water based on designed inlet temperature and flow rate to the desired outlet temperature. Instead, they are designed with an insulated stored volume capable of meeting demand at the tank capacity, a sustainable flow rate, and a recovery time. These heaters are typically installed centrally as part of a circulation loop, not at the point of use.

6.6 Subject term (key word) listing.

Hotel

Instantaneous

Semi-instantaneous

Shell

Storage

Tank

Tankless

Temperature

Thermostat

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:26Z
Check the source to verify that this is the current version before use.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4520-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:26Z
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